

Emma Ni Bhriain



In 2017 the estimated number of people living with Dementia in Ireland was 55,000 with this number expected to rise to more than 130,000 people by

2041. The purpose of my project was to create an indoor guidance system for people with Dementia. Symptoms of Dementia include memory loss and difficulty in navigating, even in familiar settings. My goal is to allow for people with mild to moderate Dementia to regain some of their independence by helping them to get from one place to another without becoming lost. The system directs a user from a starting point to a

destination of their choice using wearable devices, one on each wrist. A user can select their destination using a mobile application displayed on a tablet screen belonging to the room that they are in. The system uses Bluetooth to determine the location of the user as they move through the building. Small motors create vibrations on the wearable devices to indicate to the user whether they should turn left or right. In the case where a user does not reach

their chosen destination within a reasonable time frame, this is detected, and a notification is sent to a care assistant.

Database & Analytics	Information Systems	Game Development	Media Development	Computer Forensics	Computer Security	Software Dev: Core	Software Dev: Mobile
Automotive & Automation	Internet of Things	Cloud Computing	Computer Networks	Digital Graphic Design	Animation	Software Dev: Front End	Software Dev: Back End

Rachel Lawton



The occurrence of a stroke while driving a vehicle has the potential to cause catastrophic injuries or death to the driver or other drivers/ pedestrians.

This fourth year project is a system that will provide and maintain safety for people who unfortunately have a stroke whilst they are driving. The key to keeping the victim and surrounding drivers/ pedestrians as safe as

early detection followed by appropriate action and intervention. This project is designed along three identified cornerstones; early detection of a stroke, bringing the vehicle to a stop and informing the appropriate emergency services of the incident. This is achieved by the rapid detection of stroke symptoms through technologies such as Facial droop recognition using machine learning as well as confirmation using other IoT sensors. Such sensors include physical pressure sensors in addition to speech detection recording any abnormalities in the voice because of facial droop.

To maintain as much safety as possible for both drivers

the car at the nearest safe location is essential. To attain this the driver of the car will be notified of the closest safe place for the vehicle to pull in and come to a safe stop. If the driver is not fit for driving the vehicles Advanced Driver-Assistance System (ADAS) will take over and bring the driver to a safe stop.

It is scientifically proven that early medical intervention is key to allowing the reversal of damage caused to the victim by stroke. This is achieved by using Global Positioning System (GPS) to identify the co-ordinates of the victim and forwarding the information to the appropriate emergency services.

informing the emergency services as quickly as possible results in less permanent damage of the brain and body and the possibility of saving the person's life as well as saving the surrounding driver from potential danger.

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